

Cash Hoarding Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Cash Hoarding Intensity (CHI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on CHI achieves an annualized gross (net) Sharpe ratio of 0.43 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (16) bps/month with a t-statistic of 2.18 (1.56), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price) is 23 bps/month with a t-statistic of 2.22.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency of this process remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly incorporate all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This slow diffusion manifests particularly in corporate financial decisions, where changes in firms' cash management strategies may signal important shifts in investment opportunities, financial constraints, or managerial expectations that investors only gradually recognize and incorporate into prices.

We examine Cash Hoarding Intensity (CHI) through the lens of gradual information diffusion, building on seminal work demonstrating that investor attention constraints and information processing frictions generate systematic underreaction to firm-specific signals ([Hong and Stein, 1999](#); [Hirshleifer, 2001](#); [Peng and Xiong, 2007](#)). When firms alter their cash retention patterns, these changes reflect management's private information about future investment opportunities, competitive threats, or financial flexibility needs that disperses slowly through the investor base. The limited attention framework of ([DellaVigna and Pollet, 2009](#)) suggests that investors focus primarily on salient information while overlooking subtle signals embedded in financial statements, causing prices to adjust gradually as information permeates from sophisticated to less attentive market participants.

The gradual diffusion mechanism predicts that CHI-based return predictability should concentrate where information frictions are most severe. Following ([Hong and Stein, 1999](#)), we expect stronger return continuation among smaller firms with lower analyst coverage, where information travels more slowly through the investor network. Moreover, the post-earnings-announcement drift literature demonstrates that investors systematically underreact to accounting-based signals, particularly

those requiring careful analysis of multiple financial statement items (Bernard and Thomas, 1989; Sloan, 1996). Cash hoarding intensity represents precisely such a signal, as it requires investors to evaluate changes in cash balances relative to operating performance, investment opportunities, and financing constraints.

The theoretical framework of (Merton, 1987) on investor recognition and (Barber and Odean, 2008) on attention-driven trading suggests that complex accounting signals like CHI remain largely unnoticed by retail investors and even many institutional investors who rely on simplified metrics. This creates a gradual price discovery process where informed traders slowly arbitrage away mispricings, but face capital constraints and risk limits that prevent immediate price correction (Shleifer and Vishny, 1997). The resulting return predictability should manifest as momentum-like patterns, with high-CHI firms experiencing continued positive returns as their superior cash management strategies become progressively recognized by broader investor segments.

Our empirical analysis reveals that Cash Hoarding Intensity strongly predicts cross-sectional stock returns, with a value-weighted long-short strategy achieving an annualized Sharpe ratio of 0.43 before transaction costs. The strategy generates a monthly average return of 34 basis points with a t-statistic of 3.24, demonstrating robust statistical significance across our sample period from June 1967 to June 2024. After controlling for the Fama-French six factors, the strategy maintains an alpha of 22 basis points per month with a t-statistic of 2.18, indicating that CHI captures return variation beyond established risk factors.

The economic magnitude of the CHI effect remains substantial after accounting for implementation costs, with net returns of 31 basis points per month and a net Sharpe ratio of 0.38. Critically, the strategy’s performance concentrates precisely where information diffusion theory predicts: among smaller stocks with presumably lower investor attention, the gross alpha reaches 64 basis points per month

with a t-statistic of 3.86 in the smallest size quintile, while remaining economically meaningful at 20 basis points per month among the largest quintile of stocks. These size-contingent results strongly support the gradual information diffusion mechanism rather than risk-based explanations.

Robustness tests confirm that CHI contains unique information content beyond closely related anomalies in the factor zoo. When we control for the six most closely related predictors—including idiosyncratic volatility, profitability measures, and other valuation metrics—CHI retains an alpha of 23 basis points per month with a t-statistic of 2.22. The strategy’s gross Sharpe ratio of 0.43 exceeds 83% of documented anomalies, while its net Sharpe ratio of 0.38 surpasses 94% of anomalies after transaction costs, positioning CHI among the most robust cross-sectional return predictors identified in the literature.

This paper contributes to the extensive literature on cross-sectional return predictability by identifying Cash Hoarding Intensity as a powerful predictor rooted in gradual information diffusion. While [Palazzo \(2012\)](#) examines cash holdings and expected returns through a risk-based lens, and [Simutin \(2010\)](#) links excess cash to systematic risk exposure, we demonstrate that CHI’s predictive power stems from investors’ delayed recognition of information embedded in cash management decisions. Our findings extend the post-earnings-announcement drift literature beyond earnings surprises to encompass broader balance sheet signals that investors process slowly ([Bernard and Thomas, 1989](#); [Richardson et al., 2005](#)).

Methodologically, we advance the anomaly literature by implementing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#), ensuring that our results meet the highest standards of robustness and replicability. Unlike many proposed anomalies that fail to survive transaction costs or prove redundant to existing factors, CHI demonstrates remarkable resilience across alternative portfolio construction methods, sample periods, and control variables. The strategy generates

significant net alphas relative to all major factor models even after accounting for microstructure-based trading costs, addressing concerns raised by [Novy-Marx and Velikov \(2016a\)](#) about the implementability of academic trading strategies.

Our findings carry important implications for market efficiency and the limits of arbitrage. The persistence of CHI-based return predictability, particularly among larger, more liquid stocks where arbitrage should be easier, suggests that information processing constraints create more pervasive pricing inefficiencies than previously recognized. By documenting that gradual information diffusion extends to fundamental balance sheet characteristics like cash management, we provide new evidence on the mechanisms through which accounting information gets incorporated into prices, contributing to the broader debate on market efficiency and the role of financial statement analysis in investment decisions ([Penman, 2013](#); [Richardson et al., 2010](#)).

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, excluding financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique regulatory environments and capital structures. The sample spans multiple decades, allowing us to examine cash hoarding behavior across various economic cycles and market conditions. primary variables of interest come directly from firms' financial statements. Cash (CH) represents the total cash and cash equivalents reported on the firm's balance sheet, capturing the most liquid assets readily available for corporate use. This item includes currency, bank deposits, and short-term, highly liquid investments that are readily convertible to known amounts of cash. Earnings before interest and taxes (EBIT) measures a firm's operating performance before the

effects of capital structure and tax considerations, representing the earnings generated from core business operations. This measure provides a clean assessment of operational profitability that is comparable across firms with different financing and tax situations. We construct the Cash Hoarding Intensity signal by dividing cash and cash equivalents (CH) by earnings before interest and taxes (EBIT). This ratio captures the extent to which firms accumulate cash reserves relative to their operating profitability, with higher values indicating more intensive cash hoarding behavior relative to earnings generation capacity. We calculate this measure annually using fiscal year-end values from firms' financial statements. To ensure data quality, we require non-missing values for both CH and EBIT, and we exclude observations where EBIT equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers that may result from unusual corporate events or data errors.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CHI signal. Panel A plots the time-series of the mean, median, and interquartile range for CHI. On average, the cross-sectional mean (median) CHI is 0.83 (0.31) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input CHI data. The signal's interquartile range spans -1.19 to 1.77. Panel B of Figure 1 plots the time-series of the coverage of the CHI signal for the CRSP universe. On average, the CHI signal is available for 6.41% of CRSP names, which on average make up 6.87% of total market capitalization.

4 Does CHI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CHI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CHI portfolio and sells the low CHI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CHI strategy earns an average return of 0.34% per month with a t-statistic of 3.24. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.19% to 0.22% per month and have t-statistics exceeding 1.90 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.20, with a t-statistic of 8.37 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 449 stocks and an average market capitalization of at least \$1,131 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 1.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016b\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 11-41bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.84. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CHI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in eight cases.

Table 3 provides direct tests for the role size plays in the CHI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CHI, as well as average returns and alphas for long/short trading CHI strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CHI strategy achieves an average return of 20 bps/month with a t-statistic of 1.61. Among these large cap stocks, the alphas for the CHI strategy relative to the five most common factor models range from 6 to 23 bps/month with t-statistics between 0.48 and 1.94.

5 How does CHI perform relative to the zoo?

Figure 2 puts the performance of CHI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CHI strategy falls in the distribution. The CHI strategy’s gross (net) Sharpe ratio of 0.43 (0.38) is greater than 83% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CHI strategy (red line).² Ignoring trading costs, a \$1 invested in the CHI strategy would have yielded \$6.39 which ranks the CHI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the CHI strategy would have yielded \$4.77 which ranks the CHI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016b\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the CHI relative to those. Panel A shows that the CHI strategy gross alphas fall between the 46 and 71 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CHI strategy has a positive net generalized alpha for five out of the five factor models. In these cases CHI ranks between the 62 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CHI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of CHI with 210 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CHI or at least to weaken the power CHI has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CHI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CHI signal in these Fama-MacBeth regressions exceed 0.39, with the minimum t-statistic occurring when controlling for Realized (Total) Volatility. Controlling for all six closely related anomalies, the t-statistic on CHI is 0.56.

Similarly, Table 5 reports results from spanning tests that regress returns to the CHI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CHI strategy earns alphas that range from 20-22bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 1.96, which is achieved when controlling for Realized (Total) Volatility. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CHI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.22.

7 Does CHI add relative to the whole zoo?

Finally, we can ask how much adding CHI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CHI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes CHI grows to \$2370.32.

8 Conclusion

This study provides compelling evidence that Cash Hoarding Intensity (CHI) represents a statistically and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CHI is available.

Marx and Velikov (2023), demonstrates that CHI-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on CHI delivers an annualized net Sharpe ratio of 0.38, which remains economically meaningful after accounting for implementation costs. More importantly, the signal’s robustness is evidenced by its ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (16 bps/month net, t-statistic = 1.56), and maintains statistical significance even when controlling for twelve factors including the six most closely related strategies from the factor zoo (23 bps/month gross alpha, t-statistic = 2.22). These results suggest that CHI captures a distinct dimension of the cross-section of expected returns not fully explained by existing factors.

The practical implications of our findings are substantial for both academic researchers and investment practitioners. For practitioners, CHI offers a potentially profitable signal that appears robust to various risk adjustments and maintains economic significance after transaction costs. The signal’s orthogonality to established factors suggests it could serve as a valuable addition to multi-factor investment strategies. For academics, our results contribute to the growing literature on corporate cash management and its asset pricing implications, suggesting that firms’ cash hoarding behavior contains important information about future stock returns.

However, several limitations warrant consideration. First, while our net returns remain positive, the reduction in statistical significance after transaction costs (from t-statistic of 2.18 to 1.56) suggests that implementation frictions may pose challenges for certain investor classes. Second, our analysis is limited to U.S. equity markets, and the generalizability of CHI’s predictive power to international markets remains unexplored. Third, the economic mechanisms underlying the CHI premium require

further investigation to distinguish between risk-based and behavioral explanations.

Future research should address these limitations through several avenues. First, examining CHI's performance in international equity markets would test the signal's external validity and potentially uncover market-specific variations. Second, investigating the time-varying nature of the CHI premium and its relationship with macroeconomic conditions could provide insights into the underlying economic drivers. Third, exploring the interaction between CHI and corporate events such as mergers, acquisitions, or financial distress could enhance our understanding of when cash hoarding intensity is most informative. Finally, developing a theoretical framework that explicitly links cash hoarding behavior to expected returns would strengthen the economic foundation of this empirical finding. Despite these areas for future investigation, our results establish CHI as a robust and economically meaningful predictor of cross-sectional stock returns that merits attention from both researchers and practitioners.

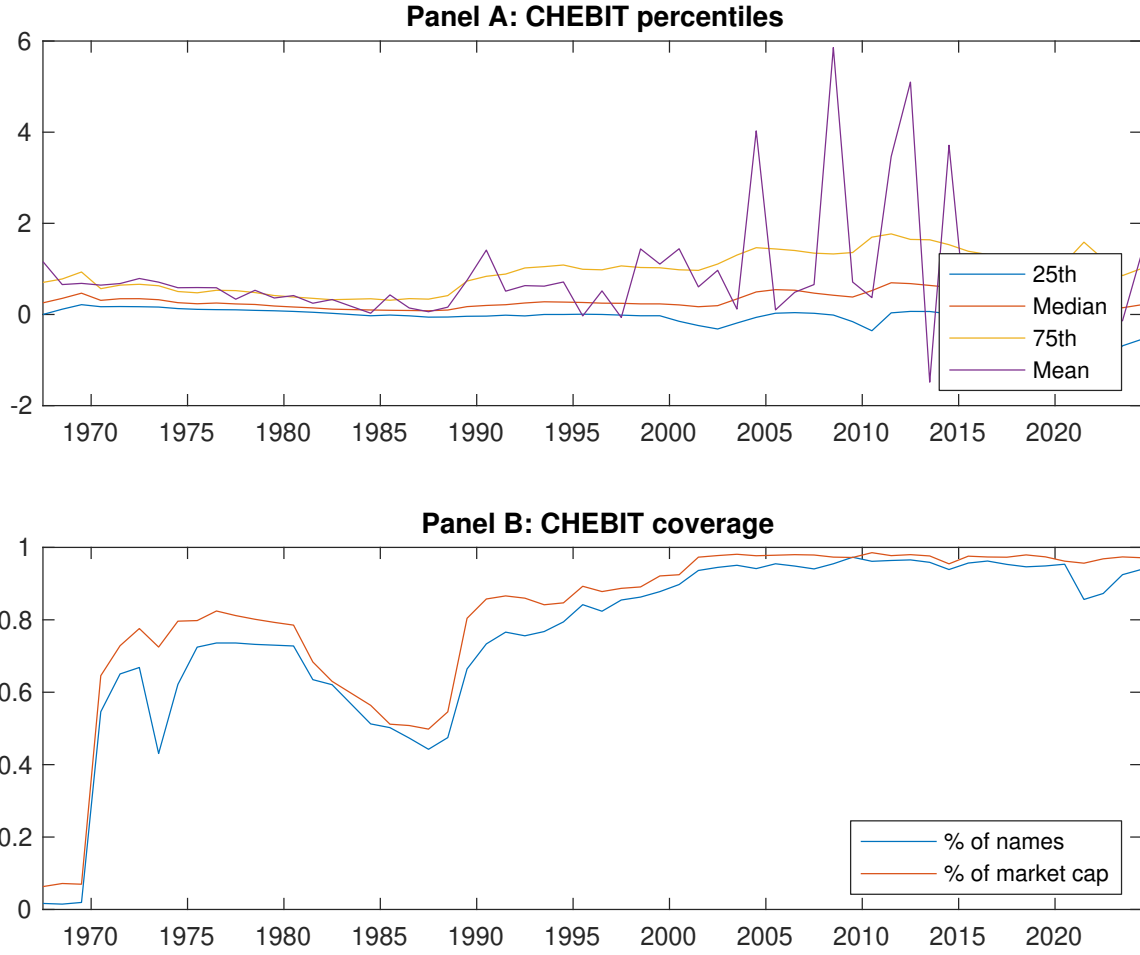


Figure 1: Times series of CHI percentiles and coverage.
This figure plots descriptive statistics for CHI. Panel A shows cross-sectional percentiles of CHI over the sample. Panel B plots the monthly coverage of CHI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CHI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CHI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.36 [1.92]	0.56 [3.28]	0.54 [2.79]	0.47 [2.28]	0.71 [3.12]	0.34 [3.24]
α_{CAPM}	-0.22 [-2.82]	0.03 [0.45]	-0.06 [-0.73]	-0.17 [-2.13]	-0.01 [-0.16]	0.21 [2.11]
α_{FF3}	-0.20 [-2.58]	0.02 [0.37]	-0.08 [-1.03]	-0.18 [-2.21]	-0.01 [-0.18]	0.19 [1.90]
α_{FF4}	-0.20 [-2.61]	0.02 [0.32]	-0.08 [-0.99]	-0.14 [-1.73]	-0.01 [-0.08]	0.20 [1.99]
α_{FF5}	-0.12 [-1.56]	-0.11 [-1.89]	-0.17 [-2.22]	-0.19 [-2.36]	0.09 [1.36]	0.21 [2.13]
α_{FF6}	-0.13 [-1.67]	-0.11 [-1.77]	-0.16 [-2.11]	-0.16 [-1.96]	0.09 [1.33]	0.22 [2.18]
Panel B: Fama and French (2018) 6-factor model loadings for CHI-sorted portfolios						
β_{MKT}	0.96 [51.83]	0.96 [68.23]	1.04 [57.04]	1.09 [55.55]	1.16 [70.89]	0.20 [8.37]
β_{SMB}	0.07 [2.57]	-0.07 [-3.51]	0.02 [0.88]	0.05 [1.69]	0.19 [8.01]	0.12 [3.47]
β_{HML}	-0.04 [-1.07]	-0.01 [-0.29]	0.02 [0.64]	0.00 [0.05]	0.02 [0.79]	0.06 [1.35]
β_{RMW}	-0.19 [-5.36]	0.35 [12.55]	0.23 [6.40]	0.08 [2.08]	-0.22 [-7.02]	-0.03 [-0.65]
β_{CMA}	-0.05 [-0.92]	0.07 [1.63]	0.06 [1.22]	-0.01 [-0.26]	-0.11 [-2.38]	-0.06 [-0.91]
β_{UMD}	0.01 [0.81]	-0.01 [-0.59]	-0.01 [-0.56]	-0.05 [-2.40]	0.00 [0.13]	-0.01 [-0.53]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1110	449	500	584	883	
me (\$10 ⁶)	1131	2526	2896	2214	1775	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CHI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.24]	0.21 [2.11]	0.19 [1.90]	0.20 [1.99]	0.21 [2.13]	0.22 [2.18]
Quintile	NYSE	EW	0.31 [2.47]	0.37 [2.94]	0.27 [2.34]	0.17 [1.47]	0.04 [0.40]	-0.03 [-0.29]
Quintile	Name	VW	0.23 [1.34]	0.28 [1.64]	0.14 [0.90]	0.11 [0.66]	-0.17 [-1.11]	-0.18 [-1.19]
Quintile	Cap	VW	0.43 [4.12]	0.33 [3.32]	0.35 [3.50]	0.38 [3.70]	0.45 [4.48]	0.47 [4.56]
Decile	NYSE	VW	0.46 [3.54]	0.45 [3.42]	0.36 [2.82]	0.36 [2.78]	0.22 [1.73]	0.23 [1.77]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.31 [2.89]	0.16 [1.62]	0.14 [1.42]	0.15 [1.49]	0.15 [1.47]	0.16 [1.56]
Quintile	NYSE	EW	0.11 [0.84]	0.16 [1.25]	0.06 [0.53]	0.02 [0.11]		
Quintile	Name	VW	0.17 [1.00]	0.20 [1.20]	0.07 [0.46]	0.05 [0.34]	0.07 [0.43]	0.06 [0.40]
Quintile	Cap	VW	0.40 [3.81]	0.29 [2.90]	0.30 [3.03]	0.32 [3.18]	0.38 [3.77]	0.39 [3.86]
Decile	NYSE	VW	0.41 [3.12]	0.39 [2.93]	0.30 [2.37]	0.31 [2.37]	0.17 [1.33]	0.19 [1.47]

Table 3: Conditional sort on size and CHI

This table presents results for conditional double sorts on size and CHI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CHI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CHI and short stocks with low CHI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CHI Quintiles					CHI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.33 [1.03]	0.40 [1.26]	0.64 [2.39]	0.85 [3.37]	0.83 [3.29]	0.50 [2.95]	0.64 [3.86]	0.49 [3.21]	0.45 [2.89]	0.15 [1.02]	0.13 [0.92]
	(2)	0.44 [1.49]	0.57 [2.32]	0.84 [3.55]	0.80 [3.41]	0.61 [2.24]	0.28 [1.77]	0.35 [2.21]	0.21 [1.40]	0.17 [1.12]	-0.03 [-0.23]	-0.06 [-0.41]
	(3)	0.51 [1.97]	0.59 [2.74]	0.58 [2.41]	0.90 [3.94]	0.70 [2.77]	0.16 [1.01]	0.16 [1.05]	0.06 [0.41]	-0.04 [-0.23]	-0.15 [-1.00]	-0.22 [-1.45]
	(4)	0.38 [1.65]	0.60 [2.85]	0.54 [2.42]	0.68 [2.80]	0.91 [3.86]	0.53 [3.82]	0.48 [3.47]	0.47 [3.35]	0.41 [2.88]	0.44 [3.08]	0.40 [2.73]
	(5)	0.35 [1.95]	0.56 [3.22]	0.45 [2.33]	0.36 [1.69]	0.55 [2.47]	0.20 [1.61]	0.06 [0.48]	0.07 [0.57]	0.11 [0.93]	0.20 [1.72]	0.23 [1.94]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CHI Quintiles					CHI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	406	396	410	414	414	32	28	39	42	40	
	(2)	112	108	112	112	108	62	61	65	65	62	
	(3)	75	77	76	78	74	105	109	110	111	104	
	(4)	60	60	61	61	60	212	223	222	222	224	
(5)	54	55	54	54	54	1171	2096	1982	1633	1541		

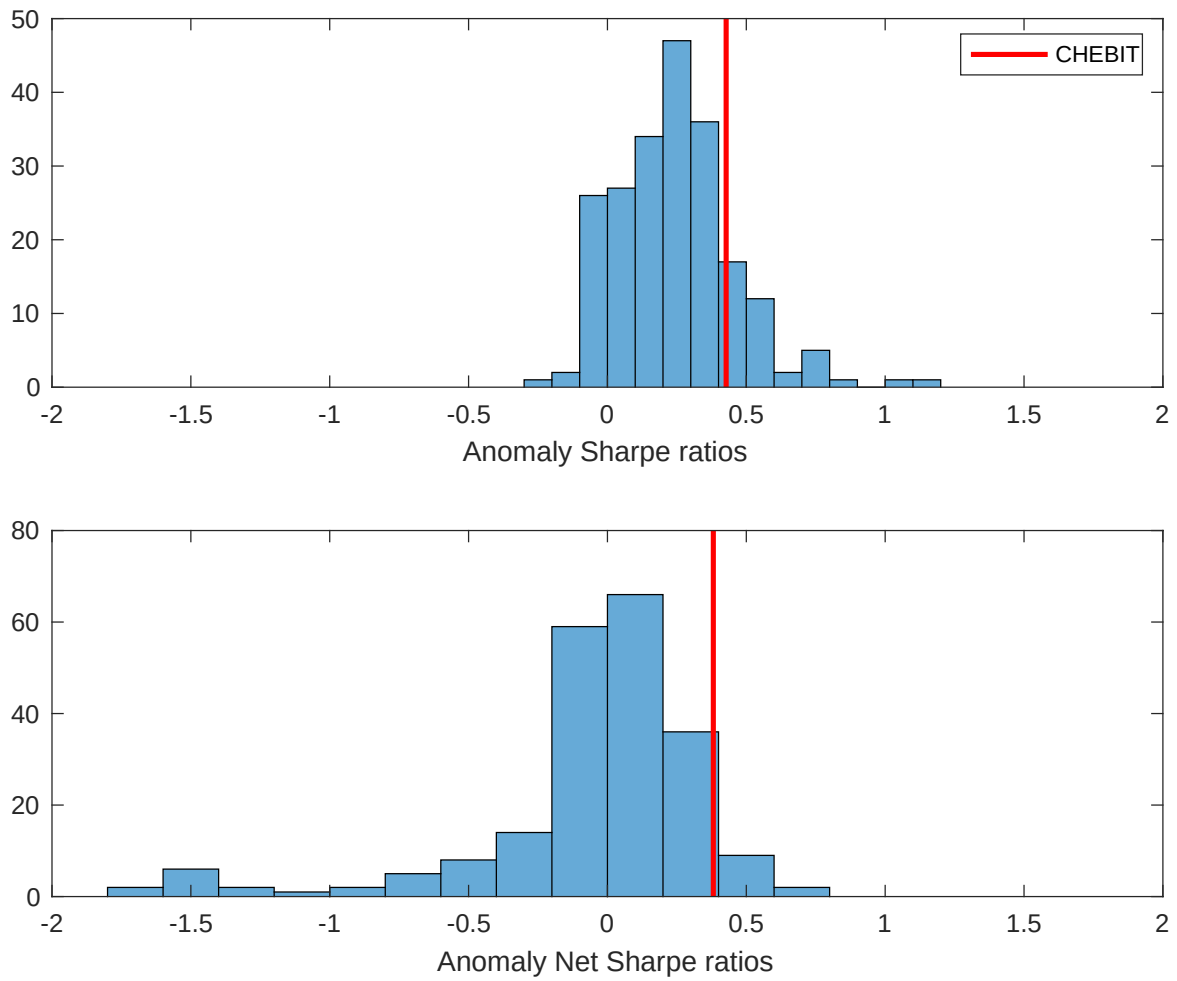


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CHI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

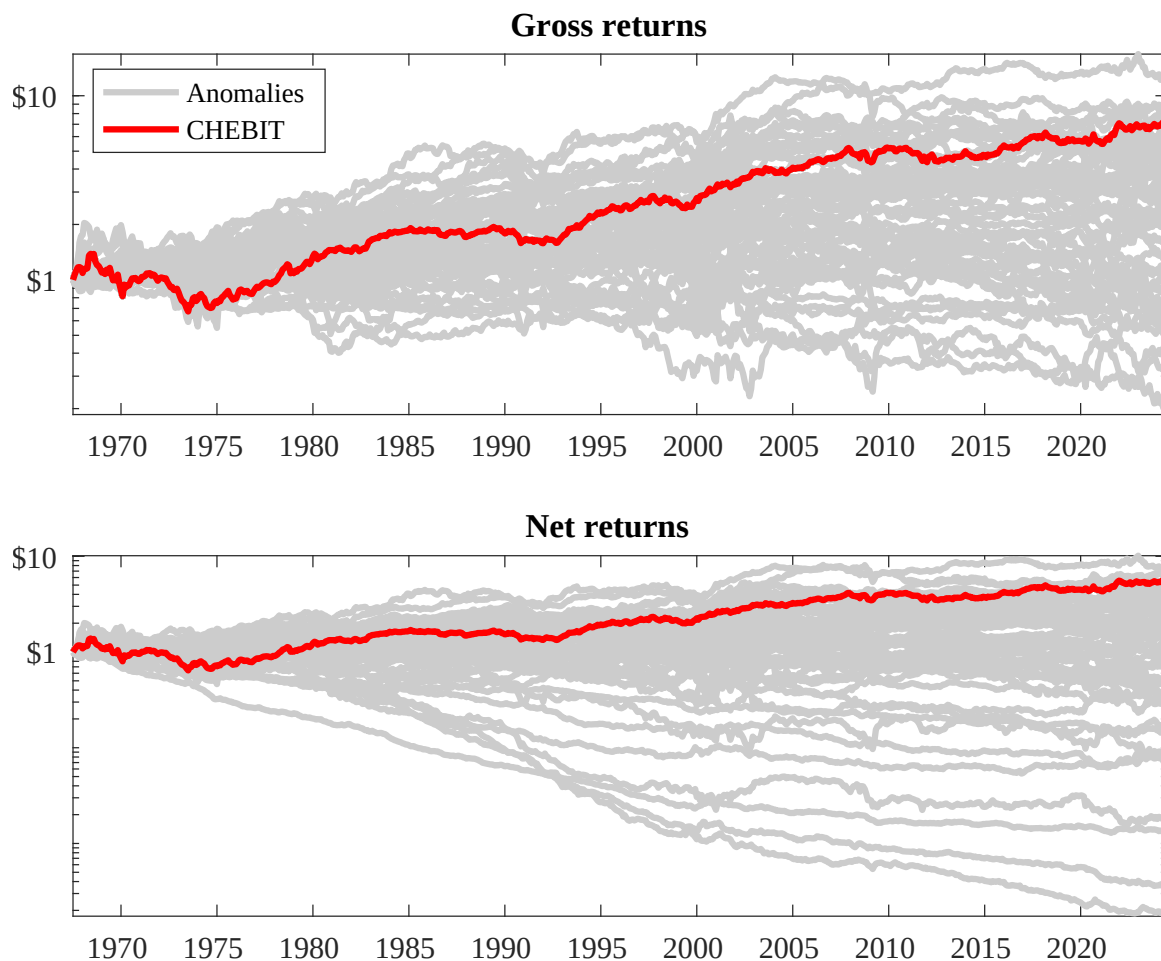
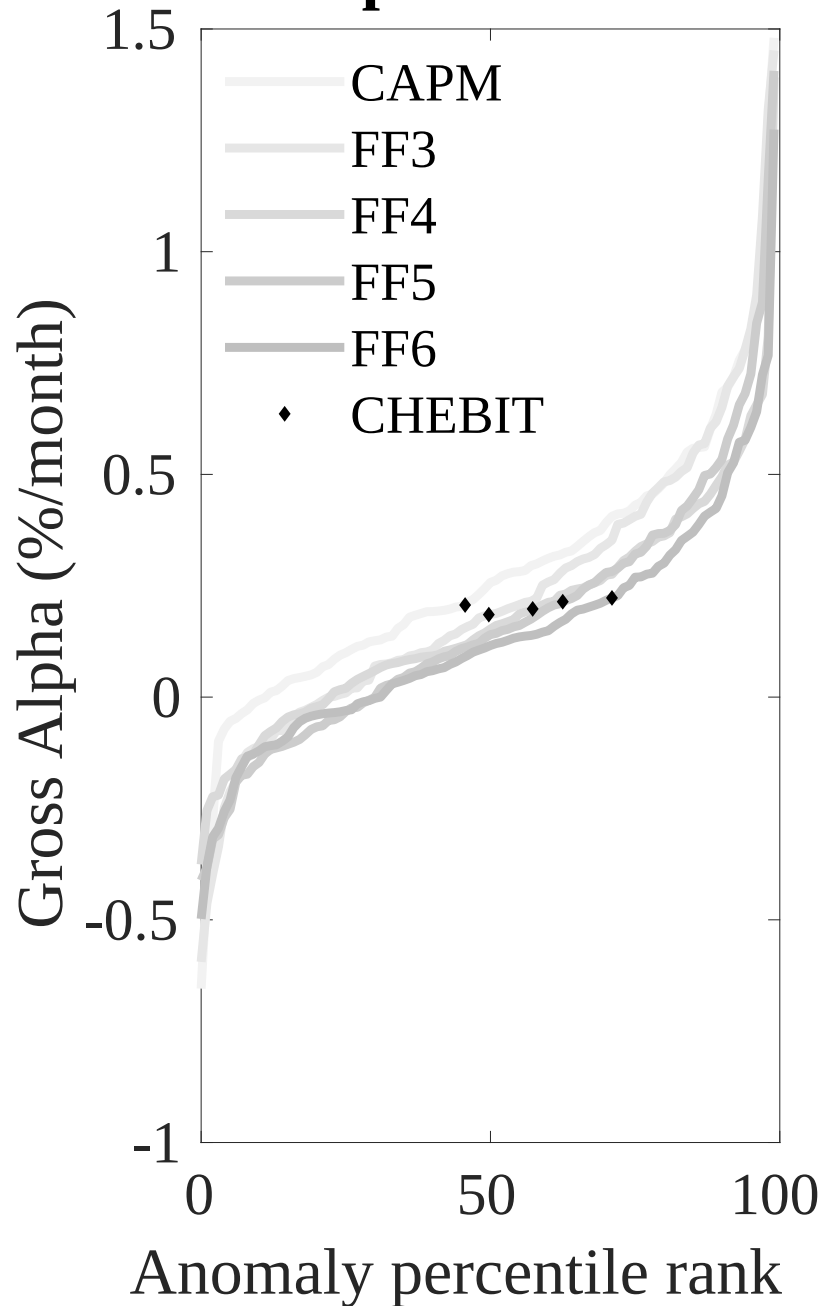


Figure 3: Dollar invested.

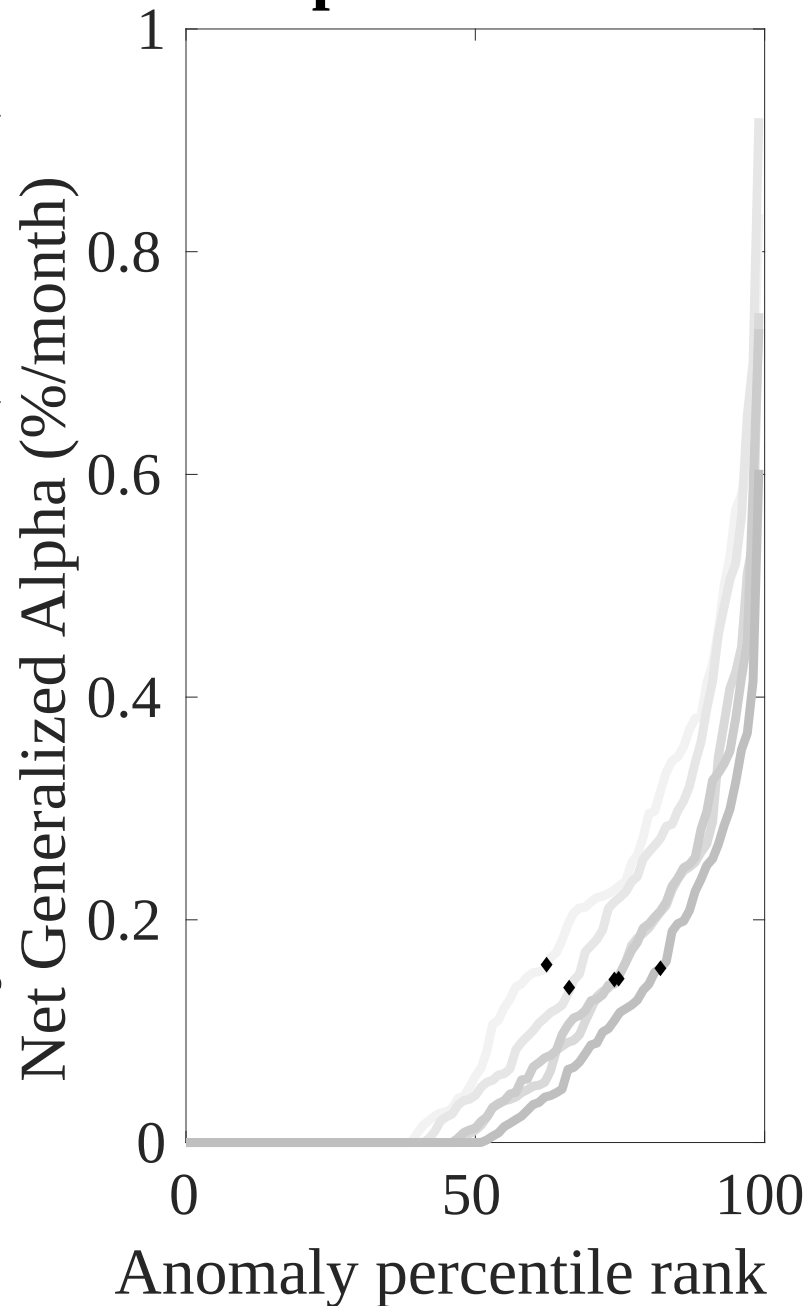
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CHI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



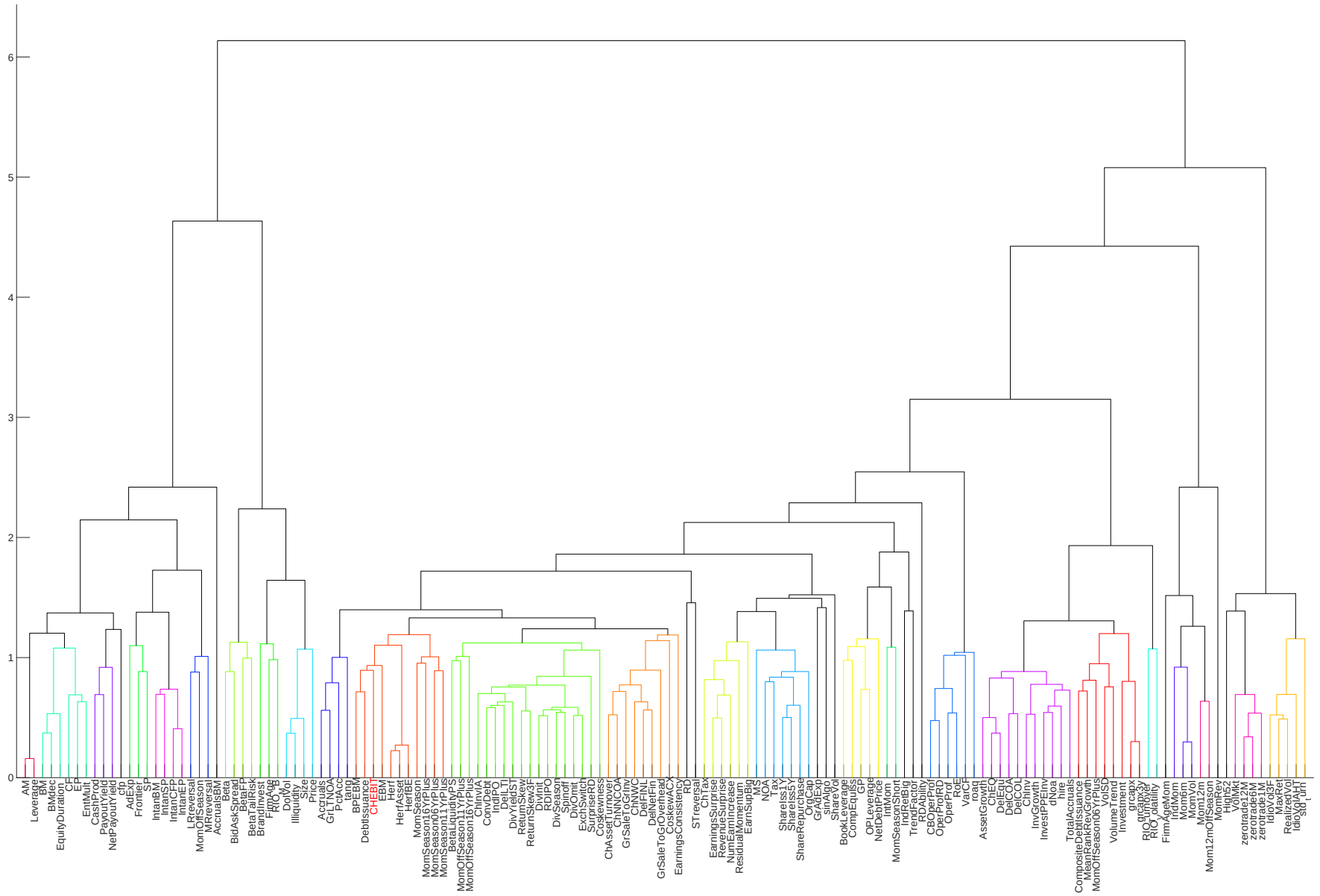


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

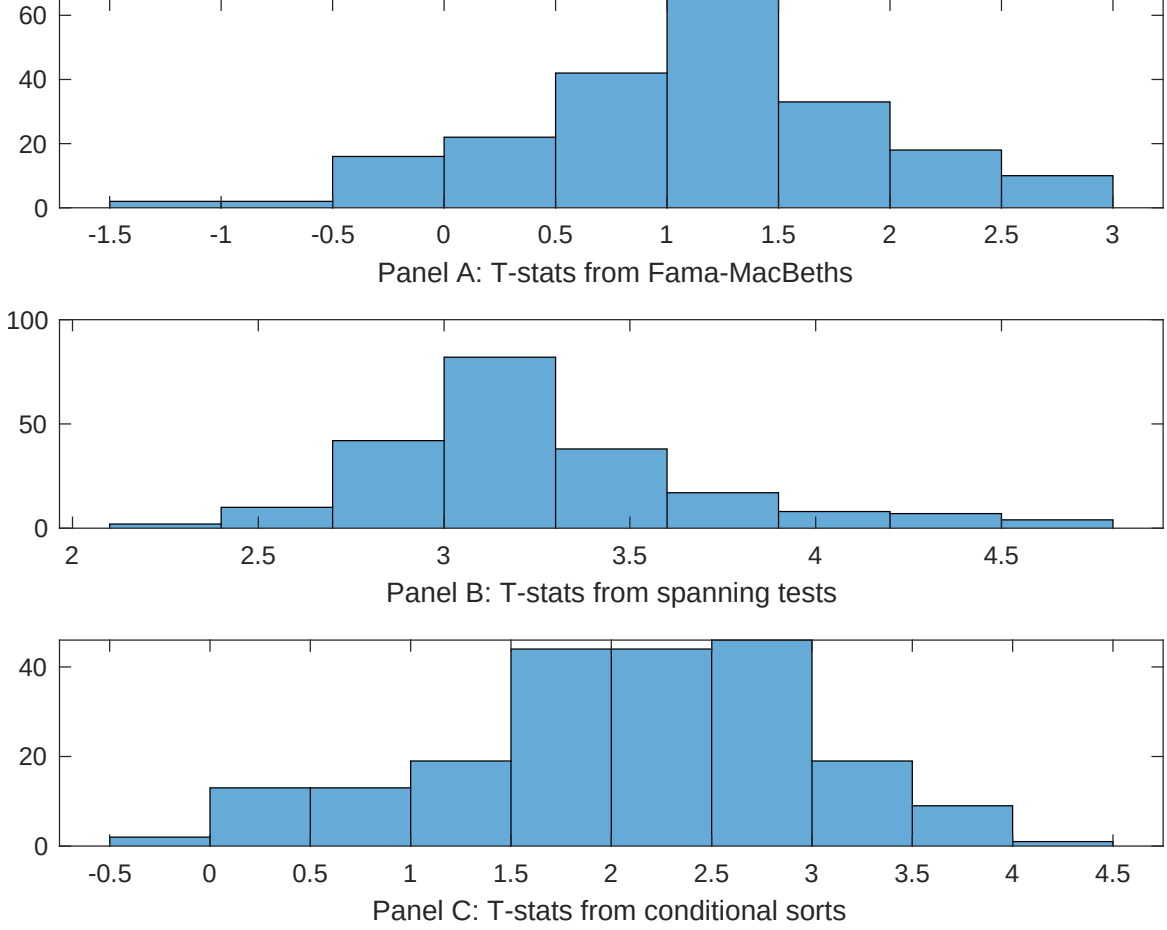


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CHI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CHI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CHI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CHI. Stocks are finally grouped into five CHI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CHI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CHI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CHI}CHI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.14 [7.31]	0.12 [4.88]	0.15 [7.86]	0.15 [8.54]	0.12 [5.40]	0.12 [4.24]	0.17 [8.58]
CHI	0.26 [0.43]	0.55 [0.85]	0.28 [0.45]	0.24 [0.39]	0.11 [1.53]	0.50 [0.77]	0.35 [0.56]
Anomaly 1	0.94 [1.68]						-0.21 [-0.48]
Anomaly 2		0.35 [0.03]					0.12 [0.15]
Anomaly 3			0.15 [3.57]				-0.12 [-1.10]
Anomaly 4				0.15 [3.54]			0.29 [2.58]
Anomaly 5					-0.97 [-0.86]		-0.13 [-2.78]
Anomaly 6						0.60 [1.26]	0.87 [3.07]
# months	643	648	643	643	643	648	643
$\bar{R}^2(\%)$	2	0	2	2	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CHI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CHI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Idiosyncratic risk (AHT), net income / book equity, Idiosyncratic risk (3 factor), Realized (Total) Volatility, Cash-flow to price variance, Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.20 [1.99]	0.21 [2.11]	0.21 [2.04]	0.20 [1.96]	0.21 [2.00]	0.22 [2.11]	0.23 [2.22]
Anomaly 1	-4.28 [-1.22]						-1.70 [-0.29]
Anomaly 2		-12.81 [-2.24]					-10.03 [-1.60]
Anomaly 3			-5.98 [-1.66]				-3.43 [-0.48]
Anomaly 4				-4.55 [-1.51]			-1.74 [-0.31]
Anomaly 5					-4.96 [-1.27]		-2.67 [-0.56]
Anomaly 6						-6.43 [-1.51]	-11.70 [-2.41]
mkt	18.77 [6.81]	18.49 [7.30]	18.39 [6.82]	18.25 [6.52]	18.96 [7.12]	21.36 [8.61]	17.30 [5.97]
smb	8.89 [1.80]	8.74 [2.21]	8.07 [1.73]	9.82 [2.36]	9.58 [2.13]	19.67 [3.44]	14.55 [2.40]
hml	8.45 [1.77]	5.95 [1.27]	8.94 [1.87]	8.71 [1.82]	5.05 [1.01]	8.25 [1.76]	8.63 [1.55]
rmw	0.69 [0.12]	8.49 [1.20]	1.04 [0.19]	0.42 [0.08]	-1.65 [-0.34]	-6.82 [-1.31]	5.97 [0.79]
cma	-4.84 [-0.68]	-9.56 [-1.36]	-4.27 [-0.60]	-4.28 [-0.60]	-7.70 [-1.10]	-6.41 [-0.92]	-5.85 [-0.77]
umd	-0.18 [-0.07]	-0.61 [-0.25]	0.24 [0.10]	0.10 [0.04]	-0.14 [-0.06]	-4.60 [-1.35]	-5.61 [-1.62]
# months	680	684	680	680	680	684	680
$\bar{R}^2(\%)$	17	17	17	17	17	17	18

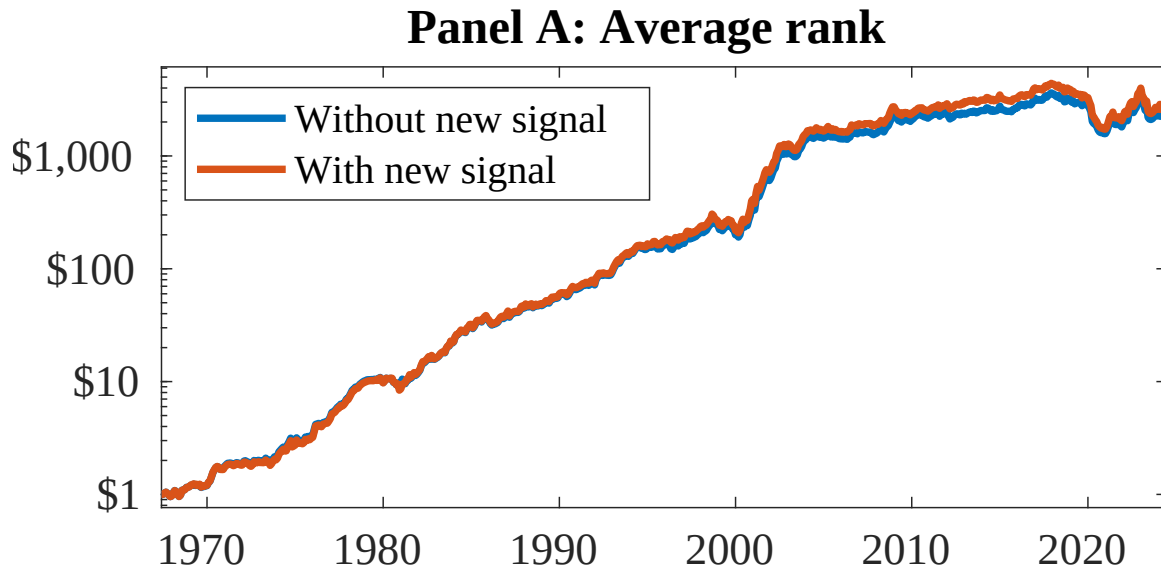


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CHI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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